fNIRS-Respiration-ECG

Published: 21 August 2024 | Version 1 | DOI: 10.17632/p74nyzt979.1

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Description

This file contains data from a single fNIRS channel (FpZ) recorded alongside ECG and respiration data during a respiratory task. Each file corresponds to an individual subject and includes an "external_sensor" section, where each row is formatted as: "number | 100 samples of ECG, 1 sample of respiration."

The respiratory task begins when the first "1" appears in the "stimulus" file. Note that the fNIRS recording starts with a 0.6-second delay, so this should be considered in your processing. The sampling frequency of fNIRS is 10 Hz, but to determine the sampling frequencies of ECG and respiration, divide the total duration by the number of samples. The sampling frequency should be approximately 6.61 Hz for respiration and 661 Hz for ECG.

ECG is recorded using a standard lead I configuration with the AD8232 bio-potential sensor, and respiration is recorded with a chest band.

Please note that some R peaks in the ECG data are corrupted due to sensor movement and noise. Therefore, complex peak detection and correction methods are required for accurate HRV extraction.

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Files

data.zip

Latest version

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Categories

Respiration, ECG Database, Functional near Infrared Spectroscopy

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